ELSEVIER

Precambrian Research 69 (1994) 51–60

Precambrian Research

Palaeomagnetism of the Mesoproterozoic Grønnedal–Íka alkaline complex, Southwest Greenland

J.D.A. Piper

Geomagnetism Laboratory, Department of Earth Sciences, University of Liverpool, P.O. Box 147, Liverpool L69 3BX, UK

Received July 9, 1992; revised version accepted January 27, 1994

Abstract

The Grønnedal–Íka complex (1299 ± 17 Ma, Rb–Sr isochron) is the earliest dated episode within the Gardar Igneous Province of South Greenland. It was intruded by NE trending dyke swarms and dislocated by faulting later in the history of the Province. A shallow 'A' magnetic remanence component present at most sites and recorded by both polarities (mean $D/I = 269/21°$, $\alpha_{95} = 5.1°$, 53 samples) is interpreted as a primary cooling-related magnetisation. It is slightly steeper than remanence in similar plutons ~10 Ma younger in the east of the Province and extends definition of a motion of the Laurentian Shield towards the palaeoequator recognised from the earliest phases of Gardar magmatism (~1310–1290 Ma). The 'A' remanence is sporadically retained in amphibolite bodies near the contact of the complex, although overprinting by the single-polarity NE-trending dyke emplacement episode between 1220 and 1160 Ma (mean remanence $D/I = 283/60°$, $\alpha_{95} = 6.1°$, 22 samples) is more prominently recorded in the country rocks. Other steep single-polarity components were probably overprinted during more than one episode in late Gardar and Mesozoic–Recent times. The palaeopole linked to the emplacement age of the Grønnedal–Íka complex (228°E, 9°N, dp/dm = 2.8/5.3°) is moved to 212°E, 13°N following adjustment for Phanerozoic drift between Greenland and North America. A westerly APW motion through ~25° is defined by plutons emplaced early in the Gardar episode (~1300–1290 Ma) and it continues with a southerly motion towards pole positions from the Mackenzie igneous episode (~193°E, 3°S) dated 1267 Ma.

1. Introduction

Following the widespread ~1950–1700 Ma mobile episodes broadly grouped as "Hudsonian", the Precambrian palaeomagnetic record from the Laurentian Shield relies mainly on data from a number of thick sedimentary successions such as the Beltian (~1600 Ma) and the Torridonian (~1100 Ma) of uncertain duration, plus a few intense igneous events. The latter include the Keweenawan (~1100 Ma) and the Franklin (~700 Ma) episodes. Recent evidence suggests that these igneous events were of very short duration (Palmer and Davis, 1987; Heaman et al., 1992). The Gardar Province of South Greenland has proved of importance for defining apparent polar wander (APW) for some 200 Ma prior to Keweenawan times (Piper, 1987, 1992, 1994) because it includes at least three episodes of magmatic activity (~1300–1280 Ma, ~1220 Ma and 1160–1125 Ma) separated by protracted periods of time. These intervals are bracketed by Rb–Sr isochron ages on a range of major central complexes (Blaxland et al., 1978) and include the age data quoted here. More limited Rb–Sr age data from the cross-cutting dolerite swarms are

0301-9268/94/$07.00 © 1994 Elsevier Science B.V. All rights reserved
SSDI 0301-9268(94)00013-H

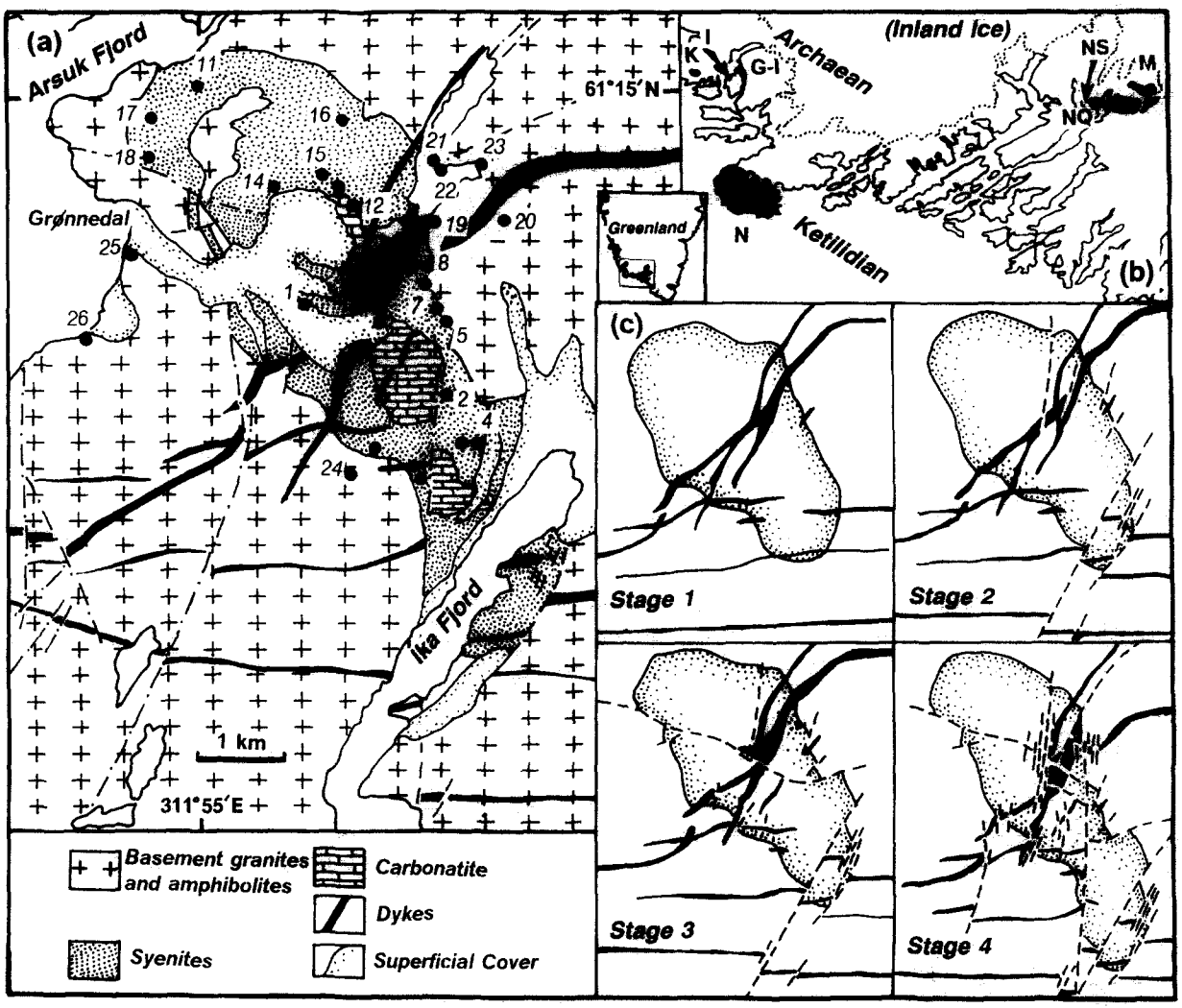


Fig. 1. (a) Geological map of the Grønnedal–Íka complex, Southwest Greenland. Palaeomagnetic sampling sites are dots with numbers. (b) Regional location of the early and mid-Gardar intrusive igneous complexes in the Gardar Province; the regional location in southern Greenland is shown in the small inset. Complexes mentioned in the text are Kûngnât (*K*), Ivigtut (*I*), North Qôroq (*NQ*), Motzfeldt (*M*), Narssarssuaq Stock (*NS*) and Grønnedal–Íka (*G–I*); *N* is the Nunarssuit complex. The dotted line is the boundary of the inland ice. (c) The sequence of late Gardar deformational events responsible for the present outcrop of the complex (after Emeleus, 1964).

given by (Patchett et al., 1978). This paper reports a palaeomagnetic study of the earliest complex yet identified within this province at Grønnedal in SW Greenland.

2. Geological outline

Regional descriptions of the Gardar Province are given by Emeleus and Upton (1976) and Upton and Emeleus (1987). The igneous activity is focussed close to the tectonic boundary between the Archaean craton of South Greenland and the Ketilidian mobile belt. The latter skirts the southern perimeter of Greenland with an approximate E–W trend (Bridgwater et al., 1973) and developed within a deeply eroded granitic basement dated 1780±20 Ma with a final cooling age of ~1600 Ma (van Breemen et al., 1974).

Early graben infill of a NE–SW-directed rift structure at ~1300 Ma was concentrated in the east of the province and was closely followed by emplacement of major alkaline bodies near the margins of the rift structure at ~1290 Ma. A single large alkaline body was emplaced at Grønnedal–Íka some 150 km to the west at about the same time. Subsequently swarms of basic dykes were intruded; the earliest members have a WNW–SSE trend, although progressive reorientation of the regional stress field during emplacement caused this trend to swing around in stages to NE–SW (Fig. 1a). In the western sector, Gardar activity terminated with emplacement of the small Kûngnât (1219±16 Ma) and Ivigtut (1222±25 Ma) complexes which cut all but a few late members of the dyke swarms (see Fig. 1b and age data summarised by Blaxland et al., 1978 and Upton and Emeleus, 1987). In the east, magmatic activity resumed later with emplacement of giant dykes and highly fractionated alkaline complexes between 1160 and 1120 Ma.

The Grønnedal–Íka complex was originally emplaced as a steep-sided ovoidal pluton with a NW–SE axis. It is about 5.9×3.5 km in size and composed predominantly of nepheline syenites intruded by a central plug of carbonatite and breccia (Callisen, 1943; Emeleus, 1964; Fig. 1a). The nepheline syenites comprise two layered series with a strong lamination showing relatively steep original dips of 30–70°. Emeleus (1964) regards these as cumulates formed by gravity settling of the major silicates and apatite onto a steeply dipping floor. The two series are separated by rafts of gneiss, and minor intrusions of microsyenite were emplaced by late stoping. The primary ovoidal form of the body was transformed into the present elongate shape (Emeleus, 1964; Fig. 1c) by the effects of transcurrent displacements on faults with N–S to NE–SW trends. These faults also displace cross-cutting dykes in this area and are attributed to crustal dislocations late in Gardar times by Emeleus and Upton (1978). With the exception of subsidiary rifting and dyke emplacement along the coastal margin, the region has been tectonically inactive since Gardar times.

Blaxland et al. (1978) derived an eleven-point Rb–Sr isochron age of 1327±17Ma from the Grønnedal–Íka body. These authors observed Sr ratios indicative of little crustal contamination and interpreted this as evidence for a common source for the fractionated facies within the pluton. This is the oldest isochron age reported from the Gardar Province and is recalculated to 1299 Ma using presently accepted decay constants (Steiger and Jäger, 1977; Upton and Emeleus, 1987).

3. Palaeomagnetic results

Palaeomagnetic samples were collected with portable coring apparatus at each of the sites shown in Fig. 1a. Typically six to nine separate short cores were drilled at each location and oriented in situ by sighting mostly on the Sun, and otherwise by magnetic bearings in conjunction with topographic sights. Eighteen sites were sampled in the various syenite facies of the complex and a further seven sites were sampled in amphibolite bodies (formerly dykes) emplaced into the Ketilidian granitic country rocks adjoining the complex. Cores were subsequently cut into 2.4 cm long cylinders for laboratory processing. The dyke swarms in this region have also been sampled and results will be reported elsewhere (see also Piper and Stearn, 1977).

Thermomagnetic determinations (saturation magnetisation, J_s, versus temperature, see Fig. 2b) identify pure magnetite as the only saturating component in the syenites with little apparent alteration on heating in air (samples 1-3 and 13-1). Natural remanent intensities (NRM's), however, vary by three orders of magnitude and some magnetically weak samples exhibit only a paramagnetic thermomagnetic signature (sample 13-3) with generation of new magnetite on cooling.

Since the syenites are visibly laminated and the amphibolites result from late regional metamorphism of Ketilidian dolerite dyke swarms, possible effects of anisotropy on magnetic remanence have been gauged by determining anisotropy of magnetic susceptibility (AMS) on small numbers of cores from each collection. The magnetic

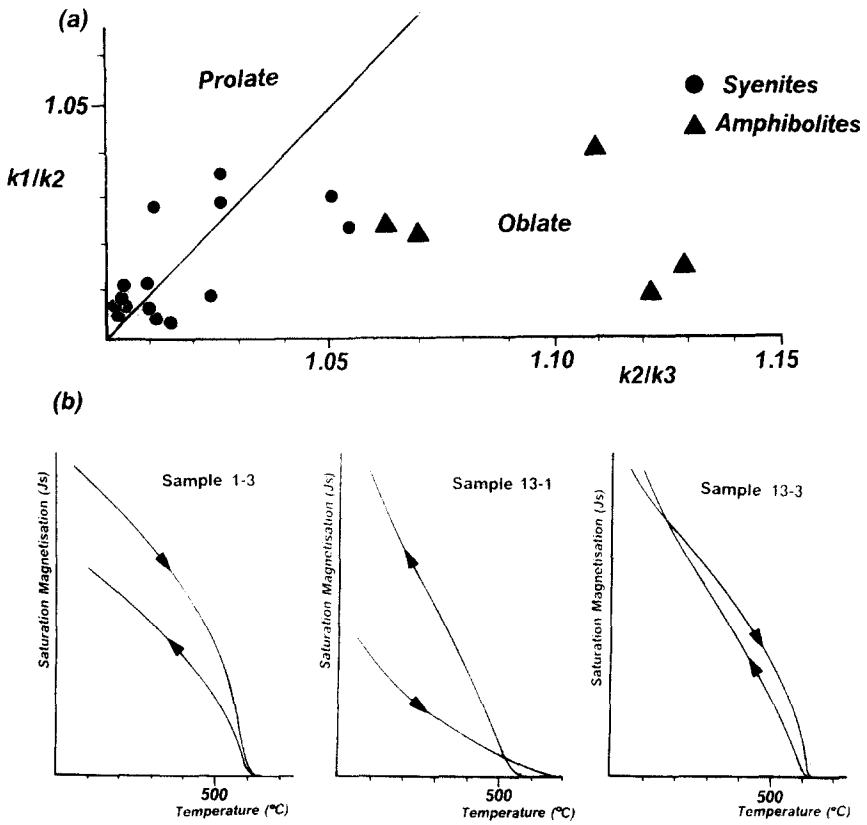


Fig. 2. (a) Anisotropy of magnetic susceptibility in samples of syenite and amphibolite illustrated as a plot of lineation parameter (k_1/k_2) against foliation parameter (k_2/k_3). (b) Thermomagnetic determinations (saturation magnetisation, J_s, versus temperature) from samples of the syenites; the J_s axis is arbitrary.

fabric is expressed in terms of an ellipsoid with maximum, intermediate and minimum axes of k_1, k_2 and k_3, respectively. The magnitude of AMS in the syenites (k_1/k_3) attains a maximum of 8% and is mostly 2% or less. Thus the visible layering in the paramagnetic silicates is not apparently reflected in the small content of magnetite responsible for the remanence and which probably dominates the measured susceptibility; thus no appreciable deflections of remanence by the effects of anisotropy are anticipated in these rocks. The magnitude of AMS is higher in the amphibolites; magnitudes of k_1/k_3 up to 16% were observed in this facies, although the likely influence on remanence directions remains small (McElhinny, 1973).

In Fig. 2a the lineation parameter k_1/k_2 is plotted against the foliation parameter k_2/k_3. It shows no preference for the small magnetic fabrics in the syenites to be either prolate or oblate. The larger anisotropies in the amphibolites, however, are oblate and are therefore likely to be dominated by paramagnetic silicates aligned in the metamorphic foliation.

Component structures of the natural remanent magnetisations (NRMs) were determined predominantly by thermal demagnetisation in steps of 50 or 100°C to 500°C, and then in steps of 15 or 20°C to Curie points of the magnetite or hematite carriers. A smaller number of cores were demagnetised in alternating fields (a.f.) in incremented steps of 5 millitesla (mT). Measurements were by "Minispin" magnetometers and demagnetisation utilised Magnetic Measurements thermal demagnetisers. Components were recognised from orthogonal vector plots devel-

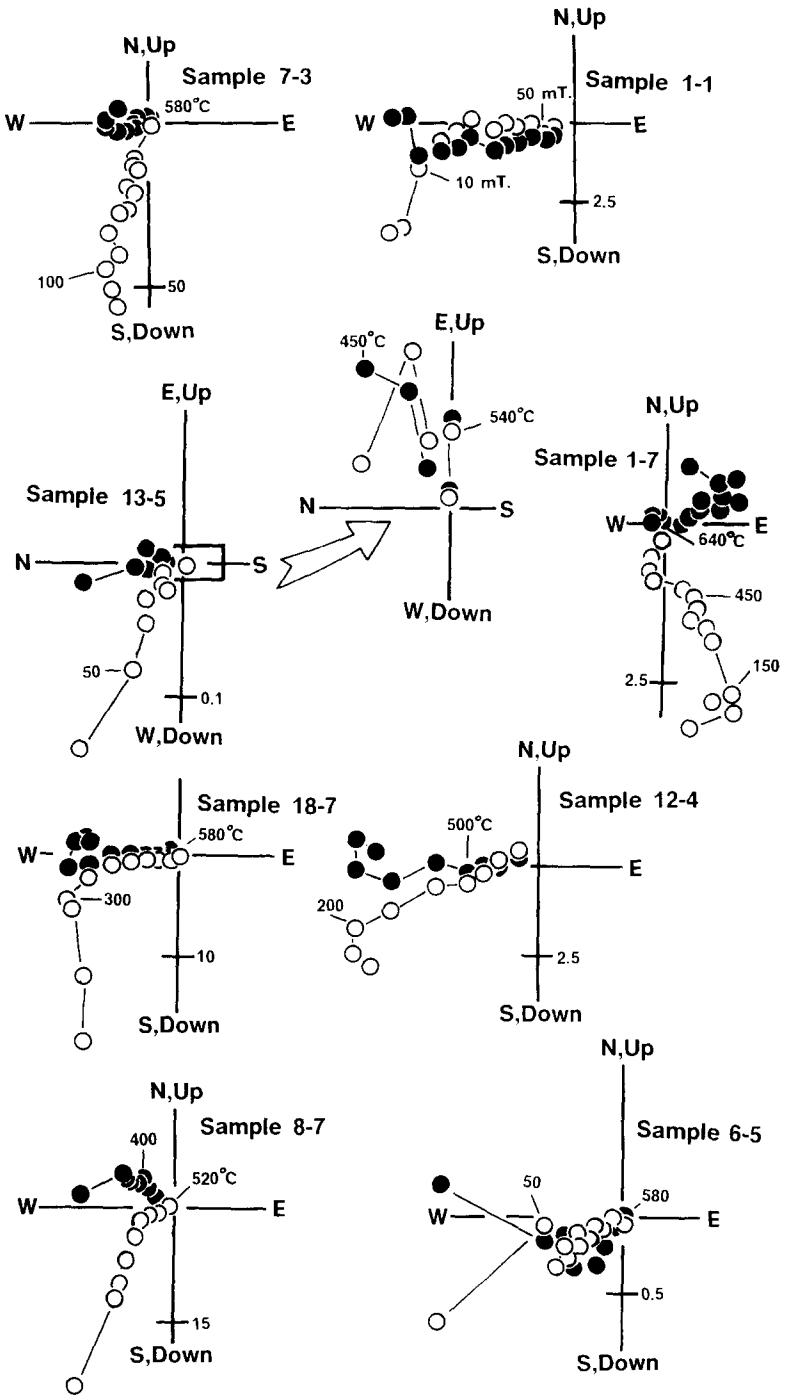


Fig. 3. Results of progressive thermal and alternating field demagnetisation of samples from the Grønnedal–Íka syenite complex. The palaeomagnetic vector is projected onto orthogonal horizontal (solid symbols) and vertical (open symbols) planes to illustrate component behaviour and the figures on the axes are intensities of magnetisation $\times 10^{-5}$ A m^2/kg. Site locations refer to Fig. 1a and Table 1.

Table 1
Site mean palaeomagnetic results, Grønnedal–Íka alkaline complex and country rock amphibolites, Southwest Greenland

Site No.	N/n	R	D	I	k	α_{95}	Component
(1) Syenites							
1	7/7	6.90	257	15	59	7.9	A
2	7/6	5.92	349	79	63	8.5	S
4	7/6	5.88	269	35	41	10.5	A
5	7/5	4.92	270	−11	51	10.9	A
7	8/4	3.99	277	78	282	5.5	S
8	8/6	5.65	301	52	14	18.2	B
9	7/3	2.96	273	12	49	17.9	A
10	8/4	3.96	347	75	80	10.3	S
12	8/5	4.92	276	25	49	11.0	A
13	7/4	3.94	269	20	50	13.1	A
14	6/6	5.95	346	74	101	6.7	S
15	9/3	2.97	278	20	66	14.5	A
	9/2	1.97	43	78	34	–	S
16	7/4	3.97	270	28	91	9.4	A
17	7/6	5.86	270	32	35	11.6	A
18	7/4	3.86	275	34	22	20.0	A
	7/2	1.98	273	73	40	–	S
(2) Amphibolites							
19	7/3	2.92	277	34	25	25.2	A
21	7/4	3.91	286	74	34	16.0	S
22	7/4	3.98	292	65	189	6.7	B
23	6/5	4.77	291	66	18	18.8	B
24	7/2	1.99	274	35	306	–	A
25	7/7	6.80	286	45	30	11.3	B
26	6/2	1.93	263	39	15	–	A
(3) Group mean results							
Grønnedal–Íka complex							
'A' component:							
53 samples		49.72	269.1	20.7	16	5.1	
(10 sites		9.71	270.6	21.2	30	8.9)	
(Palaeopole: 228.0°E, 8.9°N (dp/dm=2.8/5.3°))							
'B' component:							
13 samples		12.35	302.7	50.6	19	9.9	
(Palaeopole: 209.9°E, 42.7°N (dp/dm=8.9/13.3°))							
'S' component:							
23 samples		22.45	343.6	79.1	40	4.8	
(Palaeopole: 277.7°E, 79.7°N (dp/dm=8.8/9.2°))							
Amphibolites							
'A' component:							
7 samples		6.83	265.4	28.2	35	10.4	
(Palaeopole: 233.3°E, 10.9°N (dp/dm=6.2/11.3°))							
'B' component:							
22 samples		21.21	282.8	59.6	27	6.1	
(Palaeopole: 234.5°E, 40.5°N (dp/dm=6.9/9.2°))							

D and *I* are the mean declination and inclination of the characteristic remanent magnetisation defined by thermal and a.f. cleaning and *R* is the resultant vector derived from *n* components from a site population of *N* cores with a Fisherian precision parameter, *k*, $(=(n-1)/(n-R))$. α_{95} is the radius of the cone of 95 per cent confidence about the mean direction; dp and dm are the radii of the oval of 95 per cent confidence about the pole position in the colatitude direction, and at right angles to it, respectively. The mean location of the study area is 312°E, 61.2°N.

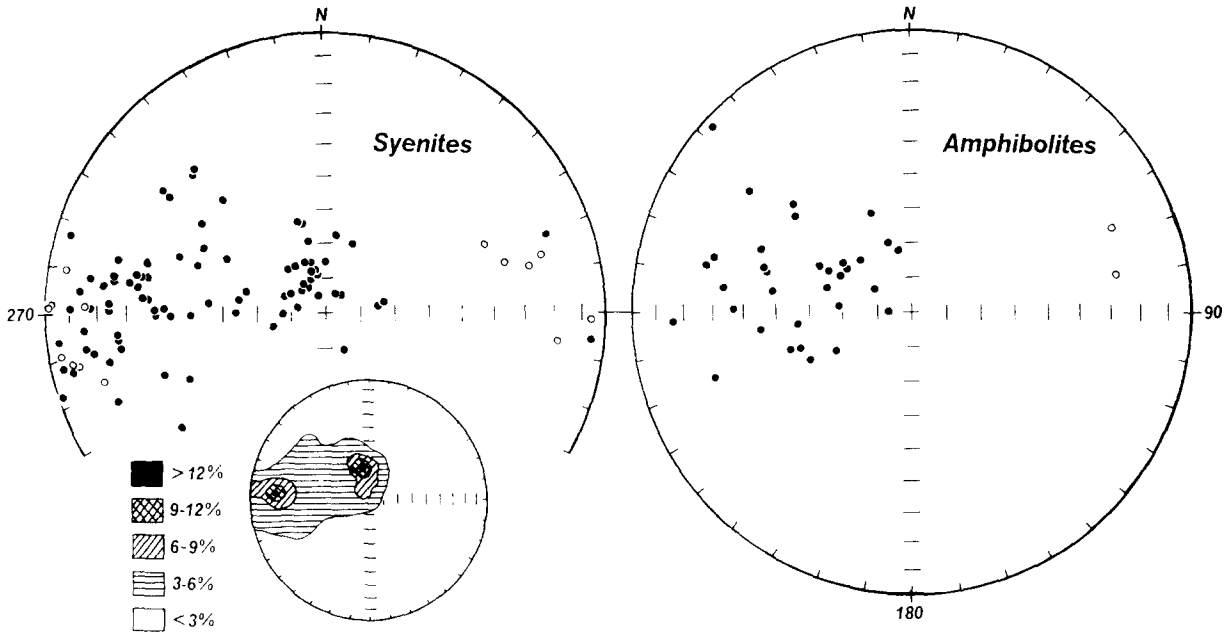


Fig. 4. Components of magnetisation derived from the syenites and amphibolites. Lower hemisphere plots are solid symbols and upper hemisphere plots are open symbols; equal area projection. The inset figure is a contoured distribution of the sample mean directions from the Grønnedal–Íka complex with all inverted (E-directed) directions reversed to a common polarity.

oped from results of the progressive demagnetisation and isolated interactively; their directions were determined by principal component analysis (Kirschvink, 1980; Sherwood, 1989).

A steep positive-directed component of magnetisation is a feature of most samples and is of composite origin. In some cases (e.g. samples 1-1, 12-4 and 18-7 in Fig. 3) it is subtracted by low alternating fields and temperatures typically below 300°C, and is evidently of viscous origin. In other samples, however, a steep remanence dominates the blocking temperature spectrum (e.g. 7-3 in Fig. 3 where two steep components constituting the remanence are evident) and in a few cases is not removed until 620–640°C and above the Curie point of magnetite (e.g. sample 1-7). This was the only hematite-held component positively identified in this study; hematite genesis by hydrothermal action (climatic-related genesis is probably excluded by the steep inclination) implies an ancient but secondary origin. It is a dominant feature at sites 2, 7, 10, and 14 (Table 1) but the mean ($D/I=344/79°$, $\alpha_{95}=4.8°$) is so close to the present Earth's field ($D/I=330/80°$) that a Recent contribution to the component structure cannot positively be resolved.

There are several possible origins for this ('S') component. The coast-bordering region of South Greenland for example, was subjected to fracturing and dyke emplacement in (?) Middle Jurassic and Late Cretaceous times accompanying rifting between Greenland and North America. The remanence in the resultant coast-parallel dykes has mean directions $D/I=336/51°$ and $330/69°$ (Piper, 1994); both polarities are present in the former population. The steeper near-vertical inclination of the 'S' component merits closer comparison with the regional field direction at ~1140–1120 Ma (ca. $D/I=324/74°$, Piper, 1977; Thomas, 1992) and corresponds to intense magmatism in the eastern part of the province towards the end of Gardar times; it is therefore suggestive of overprinting late in the cycle of igneous activity.

The shallower components of remanence are

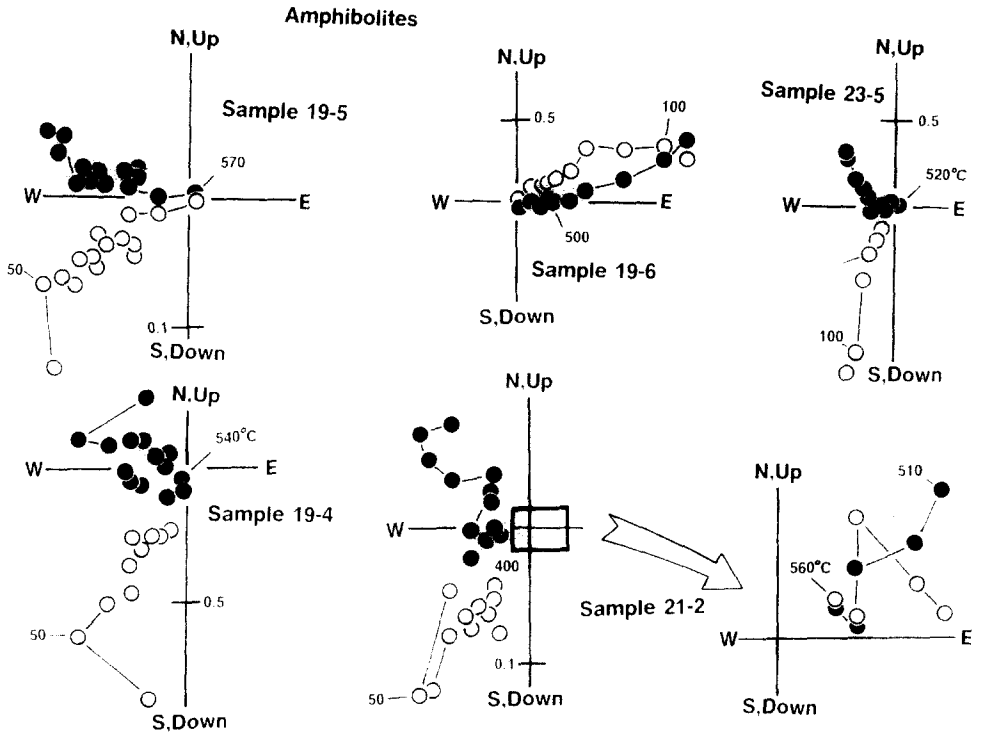


Fig. 5. Palaeomagnetic results from samples of amphibolites in terrane bordering the syenite complex. Symbols are as for Fig. 3.

directed NW positive and westerly shallow. The first ('B') component is either the dominant (13-5 in Fig. 3) or the lower blocking temperature component (sample 8-7). It is found as one polarity only and the mean ($D/I=303/51°$, $\alpha_{95}=10°$) is comparable to the remanence in the late NE-trending dykes of the Province ($D/I=290/50°$, Piper and Stearn, 1977; Piper, 1992) of which several major examples cut the complex (Fig. 1). Overprinting later in the Gardar episode of dyke injection is therefore a possible origin for these somewhat dispersed directions (Fig. 4); they are also dominant (see below) in the amphibolite country rocks.

The 'B' components are usually not quite convergent (8-7 in Fig. 3) and hence not the highest blocking temperature remanence; a small component, still with westerly declination but with shallower inclination, is recovered between ~500°C and the Curie point of magnetite. This ('A') component is the same as the single high blocking temperature component in other samples recovered by both a.f. and thermal demagnetisation (e.g. samples 1-1, 6-5, 12-4 and 18-7 in Fig. 3). There is evidence for antiparallel members in this population, either following the subtraction of a 'B' component (sample 13-5 in Fig. 3) or prior to subtraction of the west-directed 'A' remanence near the Curie point of magnetite (sample 6-5 in Fig. 3). The 'A' population comprises a near-circular group with westerly declinations and shallow, mostly positive, inclinations; ten near-antiparallel directions were also resolved and are assigned to this group (Fig. 4). The mean direction is $D/I=269/21°$ (Table 1).

This dual polarity remanence is recognised at site 19 in amphibolites near to the margin of the syenite body (samples 19-5 and 19-6 in Fig. 5). Otherwise the weaker magnetisations in the amphibolites typically have westerly declinations but steeper inclinations similar to 'B' magnetisations in the syenites (19-4 and 23-5 in Fig. 5). Locally the shallow reversed equivalent of the 'A'

direction is recovered following subtraction of 'B' (sample 21-2 in Fig. 5). Thus no clear positive contact test is available from the complex and the majority of amphibolite samples appear to have characteristic magnetisations belonging to the 'S' and 'B' populations. Unlike the syenite complex, there is no clear discrimination between the two groups in the amphibolite country rocks (Fig. 4) and group means (Table 1) are calculated arbitrarily from inclinations $>70°$ and $<70°$, respectively.

This early event in the Gardar igneous cycle was subjected to extensive later dyking and experienced profound dislocation by faulting (Fig. 1c); strike-slip motions are evident but major block rotations within the massive basement terrane (consolidated during the Ketilidian mobile episode more than 300 Ma before) are not suspected. The prolonged Gardar history was evidently responsible for the history of magnetic overprinting which is complex by comparison with the simple remanence structures in comparable, but slightly younger, complexes in the east of the igneous province (Piper, 1992). The amphibolite country rocks have proved more susceptible to overprinting than the syenites (Figs. 4 and 5). The metamorphic facies also tend to have more distributed blocking temperature spectra and yield fewer significant components.

The magnetisation record is variable within sites (Figs. 3 and 5 and Table 1); furthermore the slow acquisition of remanence within the large pluton and in the country rocks during prolonged cooling implies that individual samples will have integrated short-term variations in the magnetic field. The group means of sample populations are therefore listed in Table 1 and considered to yield representative palaeopoles. The mean derived from 'A' component site mean directions is not, however, significantly different from the sample mean (Table 1).

4. Palaeopole position

The 'A' remanence is interpreted to be cooling-related because it contrasts with younger magnetisations in the Gardar Province. This is also suggested by the occurrence of both polarities. It is dominated by west-directed magnetisations in common with plutons in the east of the province (Piper, 1992) dated ~10 Ma younger, although rather more (15%) opposite polarity components are present. The palaeolatitude ($\lambda = 11°$) is steeper and compares with $\lambda = 2°$ recorded in the Motzfeldt intrusion (1291 ± 31 Ma) and $\lambda = 1°$ recorded in the North Qôroq intrusion (1291 ± 61 Ma; Blaxland et al., 1978; Piper, 1992). A motion carrying this region towards the palaeoequator is therefore recognised at ~1300–1290 Ma (and a consequence of the APW motion shown in Fig. 6); it is continued into the opposite hemisphere by mid-Gardar (~1220 Ma) data (Piper and Stearn, 1977; Patchett et al., 1978; Piper, 1992).

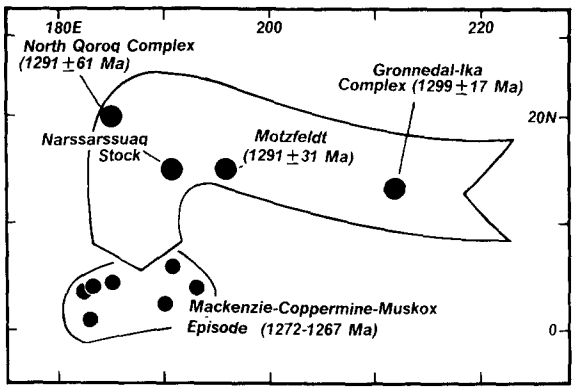


Fig. 6. Palaeomagnetic pole positions and confidence ovals derived from early Gardar alkaline complexes (this paper and Piper, 1992); the poles have been rotated into the North American reference frame and the subsequent APW motion towards the position defined by the Mackenzie igneous episode in the Canadian sector of the Laurentian Shield at 1267 Ma is indicated. For a summary of the Mackenzie poles see Piper (1988) and for a summary of Gardar palaeomagnetic results see Piper (1994).

The palaeopole position for the 'A' remanence is rotated to 212°E, 13°N with respect to North America following correction for Phanerozoic drift according to the Eulerian rotation of Bullard et al. (1965). No palaeomagnetic results from the latter region can positively be attributed to the time interval recorded by the major alkaline plutons in South Greenland (Piper, 1988). However, the well-studied Mackenzie episode is now precisely dated at 1267 ± 2 Ma

(LeCheminant and Heaman, 1989). These poles lie some 20° south of the ~1300–1290 Ma Gardar swathe and are indicative of APW motion by this amount between ~1290 and 1267 Ma (Fig. 6).

Acknowledgements

Fieldwork for this study was supported by the Natural Environment Research Council. The staff of the Ivigtut Kryolite Mine kindly provided accommodation and logistical support. I am indebted to Dr. R.L. Smith for his assistance with the field sampling, Professor B.G.J. Upton and Dr. D.N. Thomas for valuable discussions on Gardar geology and palaeomagnetism, and Mrs. J. Dean for assistance with palaeomagnetic measurements. I am grateful to the Greenland Ministry and Commission for Scientific Research in Greenland for permission to undertake these investigations.

References

Blaxland, A.B., van Breemen, O., Emeleus, C.H. and Anderson, J.G., 1978. Age and origin of the major syenite centers in the Gardar province of South Greenland: Rb–Sr studies. Geol. Soc. Am. Bull., 89: 231–244.

Bridgwater, D., Watson, J. and Windley, B.F., 1973. The Archaean craton of the North Atlantic region. Philos. Trans. R. Soc. London, A, 273: 493–512.

Bullard, E.C., Everitt, J.E. and Smith, A.G., 1965. The fit of the continents around the Atlantic. Philos. Trans. R. Soc. London, A, 258: 41–51.

Callisen, K., 1943. Igneous rocks of the Ivigtut region, Greenland, I. The nepheline syenites of the Grønne Dal–Íka area. Medd. Grønl., 131: 1–243.

Emeleus, C.H., 1964. The Grønnedal–Íka alkaline complex, South Greenland. The structure and geological history of the complex. Bull. Grønl. Geol. Unders., 85: 1–116.

Emeleus, C.H. and Upton, B.G.J., 1976. The Gardar period in Southern Greenland. In: A. Escher and W.S. Watt (Editors), The Geology of Greenland. Geological Survey of Greenland, Copenhagen, pp. 153–161.

Heaman, L.M., LeCheminant, A.N. and Rainbird, R.H., 1992. Nature and timing of Franklin igneous events, Canada: implications for a Late Proterozoic mantle plume and the break-up of Laurentia. Earth Planet. Sci. Lett., 109: 117–146.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J.R. Astron. Soc., 62: 699–718.

LeCheminant, A.N. and Heaman, L.M., 1989. Mackenzie igneous events, Canada; Middle Proterozoic hotspot magmatism associated with ocean opening. Earth Planet. Sci. Lett., 96: 38–48.

McElhinny, M.W., 1973. Palaeomagnetism and Plate Tectonics. Cambridge University Press, Cambridge, 358 pp.

Palmer, H.C. and Davis, D.W., 1987. Palaeomagnetism and U–Pb geochronology of volcanic rocks from the Michipicoten island, Lake Superior, Canada: precise calibration of the Keweenawan polar wander track. Precambrian Res., 37: 157–171.

Patchett, P.J., Bylund, G. and Upton, B.G.J., 1978. Palaeomagnetism and the Grenville orogeny, new Rb–Sr ages from dolerites in Canada and Greenland. Earth Planet. Sci. Lett., 40: 349–364.

Piper, J.D.A., 1977. Palaeomagnetism of marginal syenites and fractionated rocks of the Ilimaussaq intrusion, South Greenland. Bull. Geol. Soc. Denm., 25: 89–97.

Piper, J.D.A., 1987. Palaeomagnetism and the Continental Crust. John Wiley, New York, 434 pp.

Piper, J.D.A., 1988. Palaeomagnetic Database. John Wiley, New York, 264 pp.

Piper, J.D.A., 1992. The palaeomagnetism of major (Middle Proterozoic) igneous complexes, South Greenland and the Gardar apparent polar wander track. Precambrian Res., 54: 153–172.

Piper, J.D.A., 1994. The palaeomagnetism of Middle Proterozoic and Mesozoic dyke swarms in SW Greenland: looping in Precambrian apparent polar wander. Geophys. J. Int., in press.

Piper, J.D.A. and Stearn, J.E.F., 1977. Palaeomagnetism of the dyke swarms of the Gardar Igneous Province, South Greenland. Phys. Earth Planet. Inter., 14: 345–358.

Sherwood, G.J., 1989. Matzji—a basic program to determine palaeomagnetic remanence directions using principal component analysis. Comput. Geosci., 15: 1173–1182.

Steiger, R.H. and Jäger, E., 1977. Subcommission on Geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth Planet. Sci. Lett., 36: 359.

Thomas, D.N., 1992. Rock Magnetic and Palaeomagnetic Investigations of the Gardar Lava Succession, South Greenland. Unpubl. Ph.D. thesis, University of Liverpool.

Upton, B.G.J. and Emeleus, C.H., 1987. Mid-Proterozoic alkaline magmatism in southern Greenland: the Gardar province. In: J.G. Fitton and B.G.J. Upton (Editors), Alkaline Igneous Rocks. Geol. Soc. Spec. Publ., 30: 449–471.

van Breemen, O., Aftalion, M. and Allaart, J.H., 1974. Isotopic and geochronologic studies on granites from the Ketilidian mobile belt of South Greenland. Bull. Geol. Soc. Am., 85: 403–412